G Reproducibility and provenance

G.1 Resources, provenance, and asset scope

G.1.1 Checkpoint scope

Section 4.2 uses V-JEPA2-L for full-population Physion++ diagnostics of the specified full-clip masked-prediction score. The IntPhys2 representation study, crossed-support final256 test, post-hoc geometry transfer, and historical Kubric pilot use V-JEPA2-G. The LeWorldModel studies use separate released task checkpoints and native prediction MSE. Each experiment retains its own score definition and evidential role. The V-JEPA2-L audit is not pooled with, or treated as a substitute for, the V-JEPA2-G results. DINO-WM and the post-hoc JEPA-WM extension provide separate frozen-DINO PushT visual embedding error checks; neither score is pooled with LeWorldModel native prediction MSE.

G.1.2 Compute and software environments

The experiments ran in several environments. Table 26 maps each result to the hardware, runtime, and numerical precision used, while Table 27 reports the recorded stage times. CUDA versions are the versions reported by `nvidia-smi` or the PyTorch build suffix, as applicable.

Table 26: Compute and software environments for each experiment block.

Experiment block	Compute	Runtime, libraries, and precision
Corrected IntPhys2 representation study	NVIDIA GeForce RTX 3080 workstation; CUDA 12.8	Python 3.11.7; PyTorch 2.8.0+cu128; Transformers 5.13.1 for VideoMAE2-G; bfloat16 autocast and float16 stored features; PixelSummary uses NumPy 1.26.4.
IntPhys2 corrective probes/statistics	ARM64 Mac mini CPU; 16 GiB memory	Python 3.14.3; NumPy 2.4.3, pandas 3.0.2, and scikit-learn 1.9.0.
V-JEPA2-G multi-dose grid	NVIDIA GeForce RTX 3050 Laptop GPU; 4 GiB VRAM	Python 3.10.11; PyTorch 2.5.1+cu121; exact float32 weights with bfloat16 autocast, SDPA, and sequential transformer-block CPU/GPU offload; no quantization; unchanged 64-frame, 256-pixel preprocessing; peak reserved VRAM 704 MiB.
Physion++ static diagnostic	NVIDIA GeForce RTX 3080; CUDA 12.8	PyTorch 2.8.0+cu128 in float16; OpenCV 4.13.0, NumPy 1.26.4, pandas 2.2.1, and Matplotlib 3.10.8.
Kubric exact-rerender pilot	Cloud NVIDIA A40 (46,068 MiB); driver 570.211.01; CUDA 12.8	Rendering: Python 3.11.13, `bpy` 4.3.0, NumPy 1.26.4, Cycles OptiX, 64 samples/pixel, no denoiser. V-JEPA2-G scoring: Python 3.13.14, PyTorch 2.11.0+cu128, xFormers 0.0.35, timm 1.0.28, and einops 0.8.2; bfloat16 autocast with float32 error reduction.
LeWorldModel controls	ARM64 Mac mini CPU; macOS 26.5.2	Python 3.12.13; PyTorch 2.13.0, torchvision 0.28.0, NumPy 2.2.6, and stable-worldmodel 0.1.1; batch size 16.
DINO-WM PushT control	NVIDIA GeForce RTX 3080 workstation; CUDA 12.8	Python 3.11.7; PyTorch 2.8.0+cu128, DINO-WM repo revision `0a9492fa`, stable-worldmodel 0.1.1, NumPy 1.26.4, pandas 2.2.1; batch size 8.
JEPA-WM PushT control	Cloud NVIDIA A40 (46,068 MiB); driver 580.159.04	Python 3.10.12; PyTorch 2.7.0+cu126, torchvision 0.22.0+cu126, NumPy 2.2.6, pandas 2.3.3, Pymunk 7.3.0, JEPA-WM revision `13cf1d9c`, and stable-worldmodel revision `0ef38568`; float32, batch size 8.

The Windows workstation and cloud worker host CPU models and system memory were not captured; their accelerator models and memory were recorded. The supplementary compute record marks these fields as unavailable and gives no estimate. The exact low-memory multi-dose extraction reproduced overlapping clean features with minimum cosine 0.99999988. This engineering detail supports reproducibility and carries no scientific inference.

Table 27: Recorded wall times for the stages behind the reported results. Compute that was not recorded is excluded.

Stage	Compute	Recorded wall time
IntPhys2 development extraction	RTX 3080, CUDA 12.8	V-JEPA2-G 967.25 s; VideoMAE2-G 1296.84 s; PixelSummary 1227.36 s
IntPhys2 sealed extraction	RTX 3080, CUDA 12.8	V-JEPA2-G 1207.04 s; VideoMAE2-G 1263.22 s; PixelSummary 1769.81 s
IntPhys2 five-fold cross-fit	ARM64 Mac mini CPU	45.07 s total: development probes/statistics 17.85 s; sealed probes/statistics 27.22 s
V-JEPA2-G multi-dose grid	RTX 3050 Laptop GPU plus CPU analysis	912 clips extracted and 608 reused; extraction inference 3580.4 s; cached-feature geometry/statistics 5038.1 s
Physion++ diagnostic	RTX 3080, CUDA 12.8	48.20 s
LeWorldModel controlled tasks	ARM64 Mac mini CPU	TwoRoom 115.23 s; PushT 143.38 s
DINO-WM PushT control	RTX 3080, CUDA 12.8	143.02 s for the 256-episode controlled-displacement run
JEPA-WM PushT control	A40, PyTorch CUDA 12.6	200.90 s for 256 episodes; 13.76 s for the successful eight-episode gate
PushT role color/decomposition	ARM64 Mac mini CPU	139.12 s / 140.41 s
Kubric renderer gate	Cloud A40, CUDA 12.8	Selected/repeat 39.65/57.97 s; two denoiser trials 118.98/37.84 s
Kubric pilot render/validation/scoring	Cloud A40, CUDA 12.8	Render process 3348.14 s (72 jobs sum 3318.23 s); validation 21.16 s; scoring process 167.01 s (load 50.26 s; inference sum 35.90 s)

G.1.3 Executable provenance

The public project repository is `https://github.com/3-ml/jepa-surprise`. Code and experiment records are being prepared for release there.

The released experiment records preserve exact commands, code or file hashes, versioned configuration files, random seeds, selection manifests, output hashes, and checkpoint/data SHA-256 values. The historical anonymous supplementary archive at `https://anonymous.4open.science/r/Latent-Surprise-Audit-8730/README.md` provides source, manifests, processed metrics, integrity records, and a smoke script; the SHA-256 of `archive_manifest.json` is `56f9a8a7756806d917c075e0cb336181b8b84663689f57521773402407900b75`; third-party datasets and checkpoints are not redistributed and must be obtained under their licenses or access terms. In particular, the V-JEPA2-G and VideoMAE2-G checkpoints used for IntPhys2 hash to `67129f011434e605d894e69f2c8e13d9db118deabe59d54bf6e0fa62c2c5cb8e` and `a231169cd13307aeb29bc56259a494ef273bfd9b7232f881ac4e8d7e39713add`, respectively. The TwoRoom and PushT LeWorldModel weights hash to `566f223624ea4bfb39dbfe6ae731198dd6ea73b7b8919fed6b1ecafca810f7dd` and `48938400ae3464c9680731287f583a9cb516f55a8ec64ea13a91be47fb15b607`; the released DINO-WM PushT checkpoint hashes to `e909f0cec958fc0b49a79f2e85730ae6b5f83dee61a224705f91883eb3bb74c5`. Full package freezes are retained for the Kubric renderer and each confirmatory LeWorldModel run, and the A40 record includes the complete `nvidia-smi -q` capture. The corrected IntPhys2 Windows record has the core runtime versions but no full package freeze. It also preserves strict-load status, checkpoint hashes, feature hashes, deterministic repeat checks, and cross-model shard parity. The corrective CPU run records Python, NumPy, pandas, and scikit-learn versions plus input, code, and output SHA-256 values, but no contemporaneous complete transitive package freeze.

The post-hoc JEPA-WM record pins source commit `13cf1d9c7e476f53c17714d2e0f1dc239a883ce0`, Hugging Face revision `9b9c41ef249466630dbf1a20e78391865d07b3b9`, checkpoint SHA-256 `9beca3eafe0739c3b3adb5d734fa435ccbda0fea8a65d53d4cccec176aaaa0eb`, DINOv2 source commit `7764ea0f912e53c92e82eb78a2a1631e92725fc8`, DINOv2 weight SHA-256 `b938bf1bc15cd2ec0feacfe3a1bb553fe8ea9ca46a7e1d8d00217f29aef60cd9`, and

stable-worldmodel commit `0ef3856875e70a1283e637fcd2ab936eae6c4e6f`. The 256-episode record preserves the exact command, full package freeze, source-data and target hashes, and successful strict-load and numerical gates.

G.1.4 Asset licenses and terms of use

We use publicly released research assets, cite their original sources, and do not redistribute third-party datasets or checkpoints in this paper. Table 28 summarizes the licenses or stated access terms verified for the assets used in the reported experiments.

Table 28: Existing assets used in the experiments and their verified licenses or stated terms.

Asset	Use in this paper	License or stated terms
V-JEPA2-G checkpoint	Main IntPhys2 representation result and Kubric rerender pilot	Released Meta checkpoint; the corresponding Giant Hugging Face model card lists Apache-2.0.
V-JEPA2-L checkpoint	Local Physion++ static/surprise diagnostic	Hugging Face Large model card lists MIT.
V-JEPA2 code	Feature extraction and V-JEPA prediction utilities	Official repository is primarily MIT, with specified utility files under Apache-2.0.
VideoMAE V2 code and VideoMAE2-G checkpoint	IntPhys2 comparator	Official VideoMAE V2 code is MIT; the exact OpenGVLab/VideoMAEv2-giant Hugging Face checkpoint/model card is CC BY-NC 4.0 and is used only for noncommercial research evaluation.
IntPhys2	Evaluation videos and pair labels	CC BY-NC 4.0; stated terms restrict use to evaluation and prohibit generative uses. Our use is benchmark evaluation/probing.
LeWorldModel code, checkpoints, and task data	TwoRoom and PushT native-error controls	LeWorldModel source is MIT; the quentinll LeWM TwoRoom and PushT model/data cards list MIT.
DINO-WM code and PushT checkpoint	Matched PushT native-error control	The official DINO-WM source repository includes an MIT license; the released OSF PushT checkpoint is used only for research evaluation and is not redistributed.
JEPA-WM code and PushT checkpoint	Post-hoc matched PushT native-error control	The official JEPA-WM source and model release list CC BY-NC 4.0; the frozen DINOv2 source is Apache-2.0. Checkpoints are used only for noncommercial research evaluation and are not redistributed.
stable-worldmodel and PushT environment stack	Environment loading and PushT rendering for LeWM controls	stable-worldmodel is MIT; gym-pusht, which the PushT environment is based on, is Apache-2.0.
Kubric/MOVi-A generator	Exact rerendering pilot	Kubric code is Apache-2.0; generated MOVi-A-style pilot videos are internal evaluation artifacts.
Physion++	Local diagnostic videos in Section 4.2	The project page publicly provides download links but states a license only for the website source; no separate dataset license file was found in the local archive. We use it only for local research evaluation and cite the original work.

G.2 Study-specific reproducibility records

G.2.1 Superseded original split audit

The corrected development and sealed populations are scene-disjoint. The historical pair-held-out probe splitter, however, hashed the pair ID to assign folds and ignored the scene ID. All appearances and both pair members remained together, but other pairs from the same scene could cross folds. Table 29 reports a mechanical join of those superseded split assignments to the sample manifest. "Any overlap" counts scenes occurring in more than one fold.

A subsequent repeated scene-held-out correction kept scenes disjoint within each split but yielded 60 test scene-fold slots from only 42 unique sealed scenes: 15 scenes were never tested, while 15 recurred two or three times. Its point estimates therefore weighted scenes by test frequency, and its within-fold bootstrap did not preserve dependence across recurring scenes. Those outputs are retained only as a superseded audit trail.

Table 29: Scene overlap in the five superseded original pair-held-out probe splits.

Seed	Train	Val	Test	Tr-val	Tr-test	Val-test	Any overlap
0	46	22	18	16	8	5	29
1	51	18	23	12	22	1	35
2	50	21	21	16	16	3	35
3	45	20	21	10	12	7	29
4	47	21	20	14	12	5	31

G.2.2 Sealed-test integrity

Before creating the corrected split, we excluded the union of every inspected selection: 193 pairs across 158 scenes. The remaining Main partition was divided by scene into development and sealed test; the sealed set was exposed only after freezing the original extraction, layer, and probe specification. The five-fold cross-fit was designed after sealed-population inspection and is a post-hoc correction outside the frozen specification. The original specification SHA-256 is `f7e0ca1651c9586c180ba89c8be347d02d99de9045a8b1b9b75ed6d9fac7b465`; the sealed sample-manifest SHA-256 is `1ddc3677781e605f6d298992584768e414801dc1479b03184150dd2af11fd32a`. Both model extractors produced 912 finite shards, repeated the first sample with maximum absolute difference 0, triggered no out-of-memory fallback, and passed cross-model parity with zero missing, extra, or metadata-mismatched samples.

An initial sealed-package integrity run aborted before any representation-derived geometry metric was computed because an order-sensitive validator compared equivalent condition lists. Before any sealed geometry outcome was observed, the validator was changed to strict order-insensitive multiset equality and regression-tested; no metric or statistical code changed. Development reproduction then passed; Tables 7, 8, and 9 report the completed frozen geometry run.

Neural extraction uses bfloat16 autocast and stores float16 features.

G.2.3 Action and appearance controls

Values come from the existing `condition_summary.csv` records for the LeWM, DINO-WM, and JEPA-WM controlled-displacement studies. The original absolute contrasts and their episode-bootstrap intervals remain in Table 13; no later partial PushT continuation is used.

The released weights have SHA-256 `48938400ae3464c9680731287f583a9cb516f55a8ec64ea13a91be47fb15b607`; the evaluated dataset has SHA-256 `b6ebd9ac94bbe9e383f6e7a9cd92d74e9aa665ea57b758ed3717b0ee7df8d4fb`.

G.2.4 Physion++ scoring and validation

The pinned V-JEPA2-L model is `facebook/vjepa2-vitl-fpc64-256`, revision `b3c1679b7c34d3255ef3547f27c7b226aefab26f`, with checkpoint hash given in Appendix G.2.4.

Before new acquisition, all 337 historical scalar/available tubelet references and a first repeat matched exactly (the unchanged tolerance was 10^{-4}); all 337 canonical preprocessing tensors and ten earlier fixtures matched. Exact first-new-cell repeats and cross-study middle-window loss maps also matched. All 512-token maps are retained; appearance predictor/target vectors remain outside Git. Desktop CPU checks reconstructed every native scalar/tubelet and the appearance vector decomposition. There were no scientific retries, exclusions, or acquisition failures in these three follow-ups.

Historical Physion++ checkpoint. The historical ten-video static-content audit uses the released V-JEPA2-L checkpoint `facebook/vjepa2-vitl-fpc64-256` (checkpoint SHA-256 `25466aef85727d16546c6cf8c99f12fcfad9cbca8225d45f23685e2e025b786b`).

G.2.5 Crossed-support evaluation

The released V-JEPA2-G checkpoint has SHA-256 `67129f011434e605d894e69f2c8e13d9db118deabe59d54bf6e0fa62c2c5cb8e`. The official source revision is `204698b45b3712590f06245fbfba32d3be539812`.

The unchanged bounded context correction permits at most one uint8 step, normalized mean difference at most 10^{-6}, and changed-pixel fraction at most 10^{-4}. It canonicalizes paired context frames only in separate model-input directories: 773 cells required correction; all raw renders remain, and future bytes are unchanged. All 256 configurations passed the whole-input and visibility gates, with no excluded or missing configuration. The minimum observed target area was 385 pixels, above the 0.25% gate. Full audits covered 294,912 raw/model PNG paths, 1,024 state traces, and 512 fresh crossed trajectories.

Weights are loaded from disk one block at a time, which changes where they reside in memory but not the arithmetic. Original reference and uncached repeat scores/tubelets matched exactly.

The run was interrupted by GPU-memory and storage errors, which left partial renders and failed copies, and the scratch files for candidate 42 were missing. All failures and recoveries are recorded. The recoveries changed only job orchestration and storage. They kept the frozen protocol, the fixed candidates, the source code, and the native numerical checks, so they did not create a new experiment. Independent CPU reconstruction reproduced all 211 reported numerical statistics with a separately implemented frequency-weighted bootstrap and tie-group AUC, without neural inference. Nine unused case-colliding upstream YAML differences were disclosed; tracked executable source and the explicit constructor matched, so no blanket clean-worktree claim is made. The completion record is `results/27-19-38-wmpai-vasi-final256-complete/`.

G.2.6 Geometry-to-error feature acquisition

Checkpoint, input/context/future hashes, and disk-block target arithmetic are unchanged. Native reference/repeat, the first development feature at all five locations, and the first final feature repeat matched exactly. No new final native scores were acquired.

The complete tables retain per-world/per-appearance MSE, all predictions, and coefficient hashes.

G.3 Completion provenance and reproducibility boundaries

The 256-configuration test finished on 27 October 2026. The separately authorized three-study freeze is `b8fa8f4b23a34828df8dba6d8896f8ef3b7ee95f33f50a4e0f402f4864157809`, based on repository revision `a2c8ca864bfd916cfd7868a3dec8f9429efdaf27`. Its byte-preserved source archive and launch record precede all new cells. The 960 appearance and 20,708 static evaluations finished at 01:50 and 03:55 on 28 October; their CPU audits finished by 04:01. Geometry extraction ended at 11:36 and Desktop analysis at 11:38; Spark verification finished at 11:39 (all times +02). No further experimental acquisition was launched for manuscript preparation.

The RTX 3080 (12 GiB) ran Python 3.11.7, Torch 2.8.0+cu128, torchvision 0.23.0+cu128, Transformers 5.13.1, NumPy 1.26.4, Pillow 12.1.1, and headless OpenCV 4.13.0.92. The OpenCV/NumPy package-metadata conflict and incomplete historical Transformers provenance remain disclosed. Final rendering used the pinned Kubric/PyBullet/Blender 4.3 stack in Desktop WSL. Spark remained CPU-only (four cores, 8 GiB): it rehashed exported members/token losses/features and replayed the *same* frozen statistics implementation. A/B replay was exact; the maximum G difference was 2.22×10^{-16}, below 10^{-10}. Full B vectors were audited on Desktop, not independently on Spark. These checks differ from final256's separately implemented statistical audit. Failed Spark model and renderer gates remain failed.

Completion records are `results/28-03-55-wmpai-vasi-physion-static-complete/`, `results/28-01-50-wmpai-vasi-physion-appearance-complete/`, and `results/28-11-38-wmpai-vasi-geometry-transfer-complete/`. They retain exact commands, source/protocol/checkpoint/data hashes, seeds, complete scalar rows, predictions, summaries, failures, and external-artifact manifests. The runtime ZIP files are authoritative; newline-normalized Git text may differ from them. Large datasets, weights, canonical tensors, full vectors, and features are stored outside the repository. The anonymous supplementary archive does not yet include the

records of these experiments. A later two-model PushT continuation was not part of the approved plan and is excluded. The earlier PushT controls are unchanged.